Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/10/2005
PI Name:	Carter, Daniel C. Ph.D.		
Project Title:	Neutron Diffraction: Microgravity Applications in Structure-Guided Drug Development		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Structural biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	35824	**Congressional District:**	5
Comments:			
Project Type:	TRANSITIONAL (2004-2007)	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	09/01/2001	**End Date:**	11/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Declerq, J. P. (University of Louvain - Belgium) Evrard, C. (University of Louvain - Belgium) Myles, D. (European Molecular Biology Laboratory - c/o Institut Laue Langevin - France) Thomas, B. (AZ Technology) Bunick, G. (Oak Ridge National Laboratory) Kundrot, C. (Marshall Space Flight Center) Achari, A. (Marshall Space Flight Center)		
Grant/Contract No.:	NAG8-1828		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Since the concept was established, the DCAM has been used to produce numerous examples of macroscopic crystals of 1 cm size or greater for a variety of proteins in both flight and ground-based experiments. One of the more interesting applications of large multi-millimeter macromolecular crystals is structure determination by neutron diffraction. The International Space Station is expected to offer long duration protein crystal growth experiment opportunities to support further exploratory efforts such as: (1) Production of protein crystals of improved size and order for neutron studies in support of numerous structural biology and structure-based drug design research programs. (2) Utilization of the facility to delineate factors contributing to the effect of microgravity on the growth and quality of large protein crystals. The major objectives of this research are aimed at the production of protein crystals with sufficient size for neutron diffraction in support of research in structure-guided drug development. Extremely large protein crystals required for neutron studies are usually plagued with crystal defects causing among other things higher crystal mosaicity which can severely limit the accuracy of the final result. Large crystals grown in the diffusion-limited environment of microgravity have fewer crystal defects and represent important opportunities for neutron diffraction studies. A detailed knowledge of the atomic three-dimensional structure of protein molecules is key to a fundamental understanding of biochemistry and has important application in structure-based drug design. Neutron diffraction, with its capability of resolving hydrogen/deuterium atoms, plays a special role in macromolecular crystallography. It can be used to identify hydrogen bonds, to distinguish carbonyl oxygen and NH2 (or ND2) groups in carboxylamines, or to locate and orient water molecules, which can play key roles in protein ligand interactions and enzymatic catalysis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The DCAM (Diffusion-controlled Crystallization Apparatus for Microgravity) was installed aboard the ISS during ISS-11A within the STES 010 (STS-113 launched November 23, 2002). The STES/DCAM payload has been under active temperature control for over 900 days. This is a record for a STES payload. The hardware was originally scheduled to return in March of 2003 but currently remains on ISS under active temperature control until the resumption of the Shuttle flight program. Investigations are primarily geared toward the production of large high quality protein crystals for neutron diffraction studies and include the proteins albumin, ferritin, thaumatin, catalase, nucleosome core particle, D-xylose isomerase, bFGF, and glucocerebrosidase by investigators Dr. Daniel Carter (PI), Dr. Bill Thomas, Dr. Craig Kundrot, Dr. Aniruddha Achari, and Dr. Gerard Bunick. Large crystals of D-xylose isomerase have been produced from ground-based experiments by Gerard Bunick's group and neutron diffraction data have been collected. Their work was featured on the cover of the ACA Newsletter Number 4 in 2002. Considerable progress has been made by Dr. Daniel Carter's group at New Century Pharmaceuticals in the development of methods to produce and purify recombinant perdeuterated proteins of interest for both flight and ground-based study. Two of these perdeuterated proteins are include in those aboard the ISS. Other proteins of interest for neutron studies are being produced by recombinate methods and crystallization experiments are underway. Growing crystals large enough for neutron studies is a tedious process which requires the production of large quantities of protein. Once the conditions for crystal growth is found then many months may be required to get the crystals to grow to sufficient size for neutron diffraction.
Bibliography Type:	Description: (Last Updated: 09/11/2009)
Articles in Peer-reviewed Journals	Carter DC, Rhodes P, McRee DE, Tari LW, Dougan DR, Snell G, Abola E, Stevens RC. "Reduction in diffuso-convective disturbances in nanovolume protein crystallization experiments." Journal of Applied Crystallography 2005 Feb;38(1):87-90. http://dx.doi.org/10.1107/S0021889804028055 , Feb-2005